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Couse Name: Fundamental of data science

Case Study 1 — Employee Salaries

A company records the annual salaries (in ₹ lakhs) of 9 employees:

4, 5, 5, 6, 7, 8, 8, 9, 30

1. Find the **mean, median, and mode**.
 2. Find the **range**.
 3. Calculate the **variance and standard deviation**.
 4. Identify the outlier.
 5. Which measure of central tendency best represents the typical salary? Explain why
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1. Mean, Median and Mode

(a) Mean (Average)

Formula

$$\text{Mean} = \frac{\sum X}{n}$$

where

- $\sum X$ = Sum of all observations
- n = Number of observations

Step 1: Find the sum

Salary	Running Total
4	4
5	9
5	14
6	20
7	27
8	35
8	43

Salary	Running Total
9	52
30	82

Therefore,

$$\sum X = 82$$

Step 2: Divide by number of employees

$$\text{Mean} = \frac{82}{9} = 9.11$$

Answer

Mean = ₹9.11 lakhs

(b) Median

The median is the **middle value** after arranging data in ascending order.

The data is already arranged.

4 5 5 6 7 8 8 9 30

Since

$$n = 9$$

which is odd,

Middle position

$$\frac{n + 1}{2} = \frac{10}{2} = 5$$

The **5th value** is

7

Answer

Median = ₹7 lakhs

(c) Mode

Mode means the number occurring **most frequently**.

Salary	Frequency
4	1
5	2
6	1
7	1
8	2
9	1
30	1

The highest frequency is **2**.

Both **5** and **8** occur twice.

Answer

The data has **two modes**.

Mode = ₹5 lakhs and ₹8 lakhs

This is called **Bimodal Data**.

Case Study 2 — Website Response Time

A data scientist records the response time (in milliseconds) of a website for 10 requests:

120, 125, 128, 130, 132, 135, 140, 145, 150, 300

1. Calculate the mean and median.
 2. Find the range.
 3. Calculate the variance and standard deviation.
 4. Determine whether the data is likely to be **right-skewed or left-skewed**.
 5. Explain why the median may be more useful than the mean.
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1. Mean and Median

Mean

Formula:

$$\text{Mean} = \frac{\sum X}{n}$$
$$\sum X = 120 + 125 + 128 + 130 + 132 + 135 + 140 + 145 + 150 + 300 = 1505$$
$$\text{Mean} = \frac{1505}{10} = 150.5 \text{ ms}$$

Answer: 150.5 ms

Median

Since there are **10 values (even)**,

Median = Average of the **5th and 6th** values.

$$= \frac{132 + 135}{2} = 133.5 \text{ ms}$$

Answer: 133.5 ms

2. Range

Formula:

$$\begin{aligned}\text{Range} &= \text{Maximum} - \text{Minimum} \\ &= 300 - 120 = 180 \text{ ms}\end{aligned}$$

Answer: 180 ms

3. Variance and Standard Deviation

Using the population formula:

$$\begin{aligned}\text{Variance} &= \frac{\sum (x - \bar{x})^2}{n} = 2558.15 \\ \text{Standard Deviation} &= \sqrt{2558.15} = 50.58 \text{ ms}\end{aligned}$$

Answer:

- **Variance = 2558.15 ms²**

- **Standard Deviation = 50.58 ms**

4. Skewness

The value **300 ms** is much larger than the others, so the data is **right-skewed (positively skewed)**.

5. Why is the Median Better?

The **300 ms** value is an outlier that increases the **mean**.

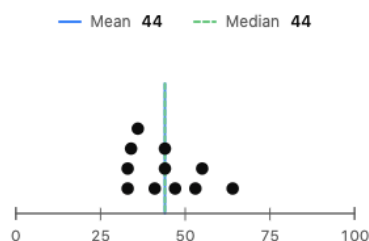
The **median (133.5 ms)** is not affected by the outlier, so it better represents the typical website response time.

Case Study 3 — Student Performance

The marks of 12 students in a Data Science test are:

42, 45, 48, 50, 52, 52, 55, 58, 60, 62, 65, 95

1. Find Q1, Q2 and Q3.
2. Calculate the **IQR**.
3. Use the **$1.5 \times \text{IQR}$ rule** to identify any outliers.
4. Find the mean and median.
5. Explain how the outlier affects the mean.

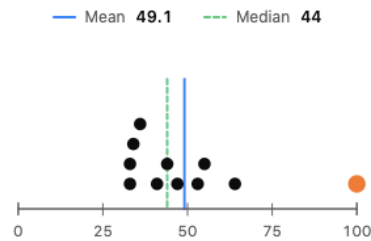


No outlier: mean is 44; median is 44

No outlier

High outlier

New data set



Outlier: mean moves 44 → 49.1; median stays 44

No outlier

High outlier

New data set

1. Find Q1, Q2 and Q3

Q2 (Median)

Since there are **12 values (even)**,

$$Q_2 = \frac{52 + 55}{2} = 53.5$$

Q1

Lower half:

42, 45, 48, 50, 52, 52

$$Q_1 = \frac{48 + 50}{2} = 49$$

Q3

Upper half:

55, 58, 60, 62, 65, 95

$$Q_3 = \frac{60 + 62}{2} = 61$$

Answers:

- Q1 = 49
- Q2 = 53.5
- Q3 = 61

2. Calculate the IQR

Formula:

$$\begin{aligned} IQR &= Q_3 - Q_1 \\ &= 61 - 49 = 12 \end{aligned}$$

Answer: IQR = 12

3. Identify the Outlier (1.5 × IQR Rule)

$$1.5 \times IQR = 1.5 \times 12 = 18$$

Lower Limit:

$$49 - 18 = 31$$

Upper Limit:

$$61 + 18 = 79$$

Any value **below 31** or **above 79** is an outlier.

Since **95 > 79**,

Outlier = 95

4. Find the Mean and Median

Mean

$$\text{Mean} = \frac{42 + 45 + 48 + 50 + 52 + 52 + 55 + 58 + 60 + 62 + 65 + 95}{12} = \frac{684}{12} = 57$$

Mean = 57

Median

$$\frac{52 + 55}{2} = 53.5$$

Median = 53.5

5. Effect of the Outlier

The score **95** is much higher than the other marks. It **increases the mean**, making it higher than the median. The **median is less affected**, so it better represents the typical student's score.

Case Study 4 — E-Commerce Orders

An online store records the number of orders received over 10 days:

18, 20, 22, 22, 24, 25, 26, 28, 30, 35

1. Calculate the mean, median, and mode.
2. Find the variance and standard deviation.
3. Calculate the coefficient of variation if the mean and standard deviation are known.
4. What does the standard deviation tell the company?
5. Would you describe the order data as highly or moderately variable?

1. Calculate the Mean, Median and Mode

Mean

Formula:

$$\text{Mean} = \frac{\sum X}{n}$$

$$\sum X = 18 + 20 + 22 + 22 + 24 + 25 + 26 + 28 + 30 + 35 = 250$$

$$\text{Mean} = \frac{250}{10} = 25$$

Mean = 25

Median

Since there are **10 values (even)**,

$$\text{Median} = \frac{24 + 25}{2} = 24.5$$

Median = 24.5

Mode

The value **22** appears **twice**, while all other values appear once.

Mode = 22

2. Find the Variance and Standard Deviation

Using the population formula:

$$\text{Variance} = \frac{\sum (x - \bar{x})^2}{n} = 22.8$$

$$\text{Standard Deviation} = \sqrt{22.8} = 4.77$$

Answer:

- **Variance = 22.8**
- **Standard Deviation = 4.77**

3. Calculate the Coefficient of Variation (CV)

Formula:

$$CV = \frac{\text{Standard Deviation}}{\text{Mean}} \times 100$$

$$= \frac{4.77}{25} \times 100 = 19.08\%$$

Answer: CV = 19.08%

4. What Does the Standard Deviation Tell the Company?

The standard deviation shows how much the daily orders vary from the average. A standard deviation of **4.77** means the daily orders usually differ by about **5 orders** from the mean.

5. Is the Data Highly or Moderately Variable?

Since the **CV is about 19%**, the variation in daily orders is **moderate**, not very high.

Answer: The order data is moderately variable.

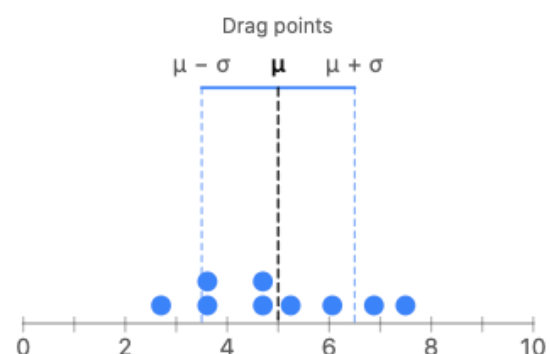
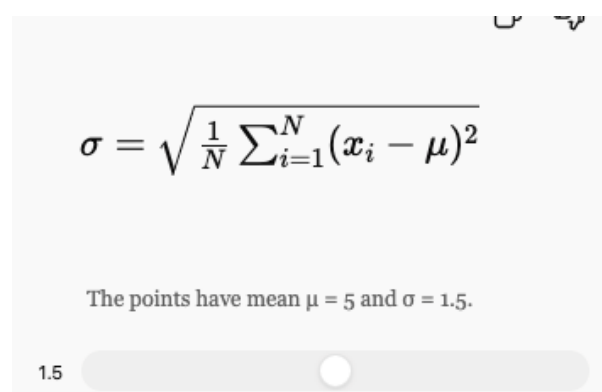
Case Study 5 — Comparing Two Data Science Models

Two machine-learning models have the following prediction errors:

Model A: 2, 3, 3, 4, 4, 5, 5, 6

Model B: 1, 1, 2, 4, 5, 7, 8, 9

1. Calculate the mean error for both models.
2. Calculate the variance for both models.
3. Calculate the standard deviation for both.
4. Which model has more consistent predictions?
5. Can the model with the lower mean error necessarily be considered better? Explain.



1. Calculate the Mean Error

Model A

Formula:

$$\text{Mean} = \frac{\sum X}{n}$$
$$\sum X = 2 + 3 + 3 + 4 + 4 + 5 + 5 + 6 = 32$$
$$\text{Mean} = \frac{32}{8} = 4$$

Mean Error (Model A) = 4

Model B

$$\sum X = 1 + 1 + 2 + 4 + 5 + 7 + 8 + 9 = 37$$
$$\text{Mean} = \frac{37}{8} = 4.625$$

Mean Error (Model B) = 4.625

2. Calculate the Variance

Using the **population variance** formula:

$$\text{Variance} = \frac{\sum (x - \bar{x})^2}{n}$$

- **Model A Variance = 1.50**
 - **Model B Variance = 8.73**
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3. Calculate the Standard Deviation

Formula:

$$\text{Standard Deviation} = \sqrt{\text{Variance}}$$

- **Model A: $\sqrt{1.50} = 1.22$**
- **Model B: $\sqrt{8.73} = 2.95$**

Answers:

- **Model A SD = 1.22**
 - **Model B SD = 2.95**
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4. Which Model Has More Consistent Predictions?

A **lower standard deviation** means the prediction errors are more consistent.

- **Model A SD = 1.22**
- **Model B SD = 2.95**

Answer: Model A has more consistent predictions because its errors vary less.

5. Can the Model with the Lower Mean Error Always Be Considered Better?

No. A lower mean error alone does not guarantee a better model. We should also consider:

- Variance
- Standard deviation
- Accuracy on new data
- Overall consistency

A model with **consistent predictions** is often more reliable.